

Seasonal Trends, Pollutant Co-variation Patterns and Data-Driven Prediction of PM_{2.5} in Delhi (2015–2020)

Mantasha Siddiqui | B.Tech. Civil Engineering, National Institute of Technology Srinagar

Technical report | Code and data: https://github.com/joya0011/Delhi_Air_quality_analysis

Abstract

Daily city-averaged CPCB monitoring data for Delhi (1 January 2015 – 1 July 2020; 2,009 days, 11 pollutants) were analysed to characterise PM_{2.5} seasonality, compliance with the Indian 24-hour NAAQS (60 µg/m³), pollutant co-variation, and predictability. PM_{2.5} exceeded the NAAQS on 84% of days in 2015 and 67% in 2019, with November–January monthly means routinely above 180 µg/m³. The PM_{2.5}/PM₁₀ ratio was markedly higher in winter (0.61) and post-monsoon (0.57) than in the pre-monsoon (0.44) and monsoon (0.42) seasons, consistent with a shift from dust-dominated to combustion-dominated aerosol. Principal component analysis identified a dominant common factor (49% of variance) loading on particulate matter, NO_x and benzene, plus a cold-season CO factor and a separate toluene factor. During the COVID-19 lockdown (25 March – 31 May 2020), NO fell by 76% and PM_{2.5} by 52% relative to 2017–2019, while O₃ was nearly unchanged (-2%). A random forest trained on 2015–2018 and tested on 2019–2020 predicted daily PM_{2.5} with R² = 0.79 using only gaseous pollutants and seasonality (vs. R² = 0.41 for a monthly climatology baseline), with benzene the most informative gaseous predictor. The model over-predicted PM_{2.5} by about 50% during lockdown, indicating that relationships learned under normal emissions did not hold when anthropogenic activity changed abruptly.

1. Introduction

Delhi experiences some of the highest fine-particulate (PM_{2.5}) concentrations of any major city, with a pronounced winter peak attributed to a combination of local combustion sources (vehicles, industry, residential and waste burning), regional crop-residue burning in the post-monsoon period, and meteorology (low wind speeds and shallow boundary layers that trap pollutants). Designing effective control measures requires understanding *when* concentrations are highest and *which* sources contribute – the goal of source apportionment and emission-inventory studies conducted for Indian cities under the National Clean Air Programme.

Formal source apportionment relies on chemically speciated measurements and receptor models such as Positive Matrix Factorization (PMF) (Paatero & Tapper, 1994). Routinely reported regulatory data (criteria pollutants from continuous monitoring stations) are far more abundant but carry less chemical information. This study asks how much can be learned about the timing, likely drivers and predictability of Delhi's PM_{2.5} from this routinely available data alone. The specific objectives were to:

- quantify PM_{2.5} seasonality and NAAQS exceedance frequency, 2015–2020;
- use the PM_{2.5}/PM₁₀ ratio, pollutant correlations and principal component analysis (PCA) to identify co-variation patterns indicative of source types;
- use the COVID-19 lockdown as a natural experiment on anthropogenic emissions; and
- evaluate how well daily PM_{2.5} can be predicted from co-measured pollutants, and which pollutants carry the most information.

2. Data and methods

2.1 Data

Daily city-level concentrations for Delhi were taken from the public *Air Quality Data in India (2015–2020)* compilation (Rao, 2020), which aggregates CPCB continuous ambient air quality monitoring data. Variables used: $PM_{2.5}$, PM_{10} , NO, NO_2 , NO_x , NH_3 , SO_2 , O_3 , benzene, toluene ($\mu g/m^3$) and CO (mg/m^3). The record contains 2,009 daily values.

2.2 Quality control and pre-processing

- Xylene was excluded (39% missing).
- Exact zero values in CO (87 days), toluene (108), benzene (8) and NO_x (2) were treated as sensor dropouts, since zero daily means are physically implausible in Delhi.
- Gaps of up to 7 days were filled by time interpolation; longer gaps were left missing.
- 16 days (0.8%) on which $PM_{2.5}$ exceeded PM_{10} – physically impossible, since $PM_{2.5}$ is a subset of PM_{10} – were excluded from the ratio analysis. These most likely arise because city averages for the two variables are built from different sets of reporting stations.
- Seasons follow India Meteorological Department (IMD) definitions: winter (Jan–Feb), pre-monsoon (Mar–May), monsoon (Jun–Sep), post-monsoon (Oct–Dec).

2.3 Analysis

Exceedance was computed against the NAAQS 24-hour $PM_{2.5}$ limit of $60 \mu g/m^3$ (CPCB, 2009) as the share of monitored days above the limit. **Co-variation** was assessed with Spearman rank correlations (robust to skewed concentration distributions) and PCA on log-transformed, standardised data for ten pollutants (NO_x excluded because it is approximately $NO + NO_2$); components with eigenvalue > 1 were retained (Kaiser criterion) and loadings are reported as pollutant–component correlations. The **lockdown effect** was computed as the percentage difference between mean concentrations for 25 March–31 May 2020 and the mean of the same window in 2017, 2018 and 2019.

For **prediction**, linear regression and random forest (Breiman, 2001; 400 trees, minimum 3 samples per leaf) models were trained on 2015–2018 and tested on 2019–2020. A chronological split was used deliberately: consecutive days are strongly autocorrelated, so a random split would leak information and overstate skill. Two feature sets were compared: **(A)** all co-pollutants including PM_{10} (an easy case, since PM_{10} contains $PM_{2.5}$) and **(B)** gaseous pollutants only, which tests whether gas-phase tracers explain particulate levels. Both include sine/cosine of day-of-year to encode seasonality. A monthly-climatology baseline (training-period mean $PM_{2.5}$ for each calendar month) shows the skill added beyond seasonality alone. Feature relevance was measured by permutation importance on the test set (20 repeats). All analysis was performed in Python (pandas, scikit-learn, matplotlib/seaborn).

3. Results

3.1 Seasonality and NAAQS exceedance

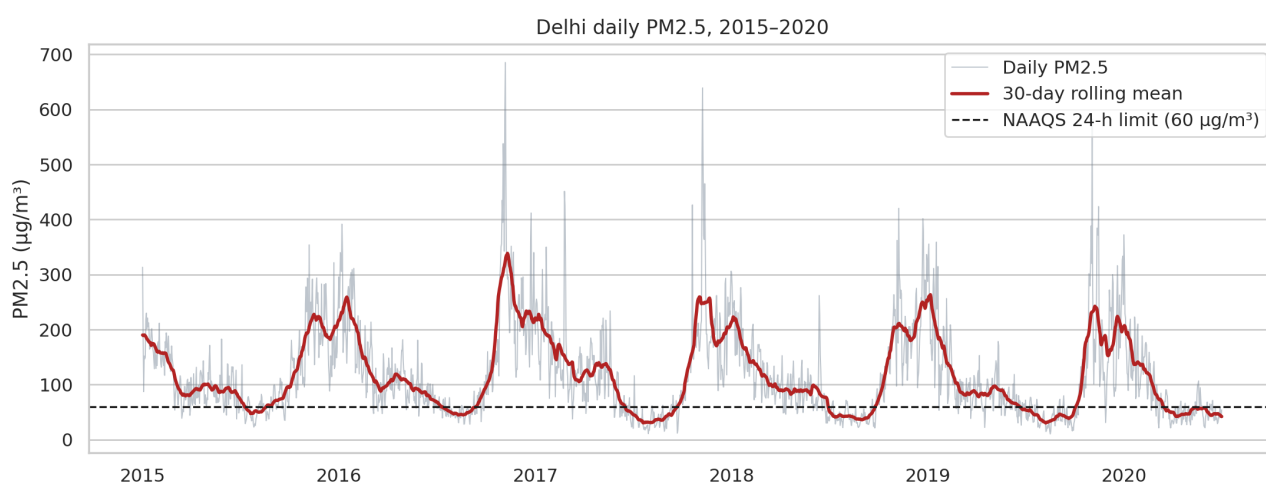


Figure 1. Daily $PM_{2.5}$ in Delhi with 30-day rolling mean and the NAAQS 24-hour limit.

$PM_{2.5}$ follows a strongly repeatable annual cycle (Figures 1–2). Concentrations rise sharply in October, peak in November–January, and fall to their lowest levels in July–September when monsoon rainfall scavenges particles and mixing is deeper. The highest monthly mean in the record was $306 \mu\text{g}/\text{m}^3$ (November 2016), about five times the NAAQS limit. Even the cleanest months (July–August) typically sit only just below the limit.

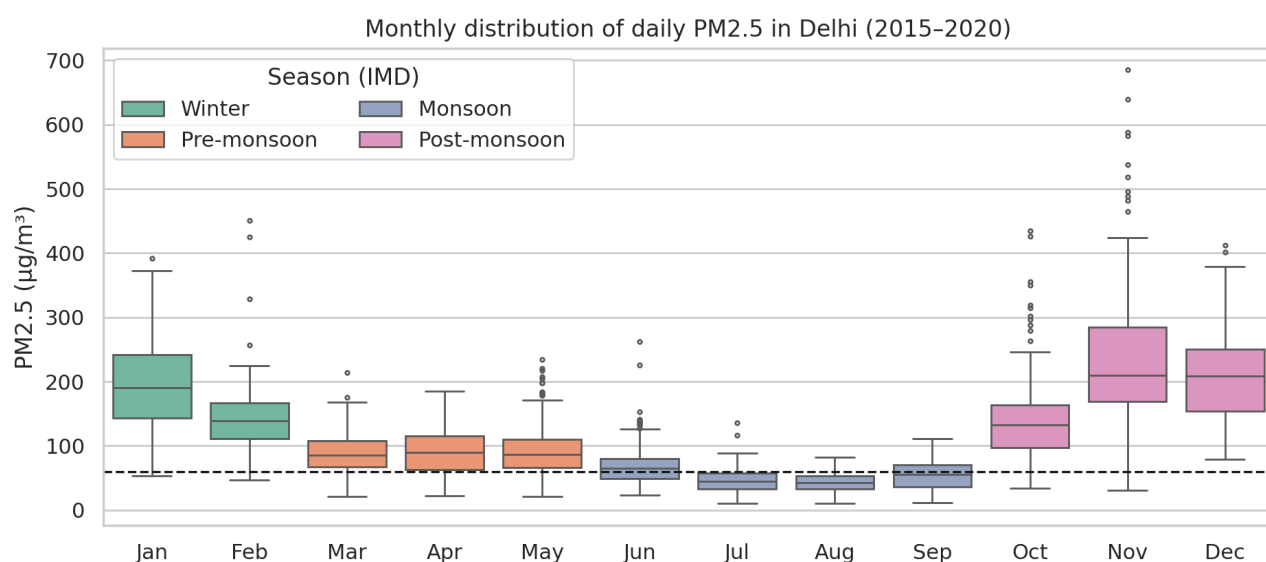


Figure 2. Distribution of daily $PM_{2.5}$ by calendar month, coloured by IMD season. Dashed line: NAAQS 24-hour limit.

Table 1. $PM_{2.5}$ NAAQS exceedance by year.

Year	Days monitored	Days > $60 \mu\text{g}/\text{m}^3$	% of days	Mean $PM_{2.5} (\mu\text{g}/\text{m}^3)$
2015	365	307	84.1	117.3
2016	366	284	77.6	138.5
2017	365	267	73.2	124.5
2018	365	265	72.6	115.0
2019	365	243	66.6	108.5
2020 (Jan–Jun)	183	90	49.2	80.3

The share of exceedance days fell from 84% in 2015 to 67% in 2019, and the annual mean fell from a peak of $138.5 \mu\text{g}/\text{m}^3$ (2016) to $108.5 \mu\text{g}/\text{m}^3$ (2019), still about 2.7 times the annual NAAQS of $40 \mu\text{g}/\text{m}^3$. This decline is descriptive only: five years is a short record, inter-annual meteorology varies strongly, and

no formal trend test was applied. A national analysis of 2015–2019 CPCB data found no statistically significant trend across India (Sharma & Mauzerall, 2022). The 2020 value covers January–June only and includes the lockdown, so it is not comparable.

3.2 $PM_{2.5}/PM_{10}$ ratio

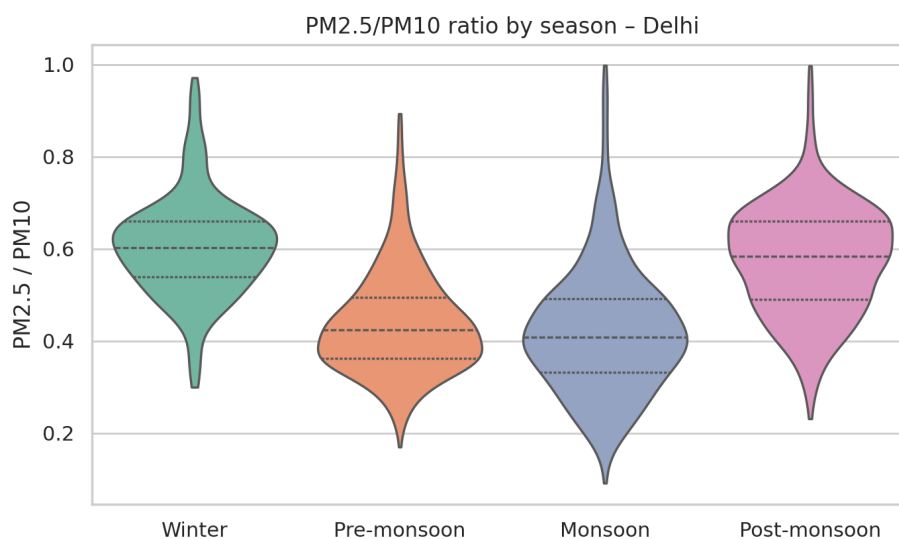


Figure 3. $PM_{2.5}/PM_{10}$ ratio by season (days with ratio > 1 excluded as physically impossible).

The fine fraction was clearly higher in winter (mean 0.61) and post-monsoon (0.57) than in pre-monsoon (0.44) and monsoon (0.42). Since coarse particles come mainly from mechanical sources (road and construction dust, wind-blown soil) while fine particles come mainly from combustion and secondary formation, this indicates that the cold-season pollution episodes are driven primarily by combustion-related and secondary aerosol rather than dust. In the pre-monsoon, higher winds and dry soils make coarse dust relatively more important, even though total $PM_{2.5}$ is lower.

3.3 Pollutant co-variation and PCA

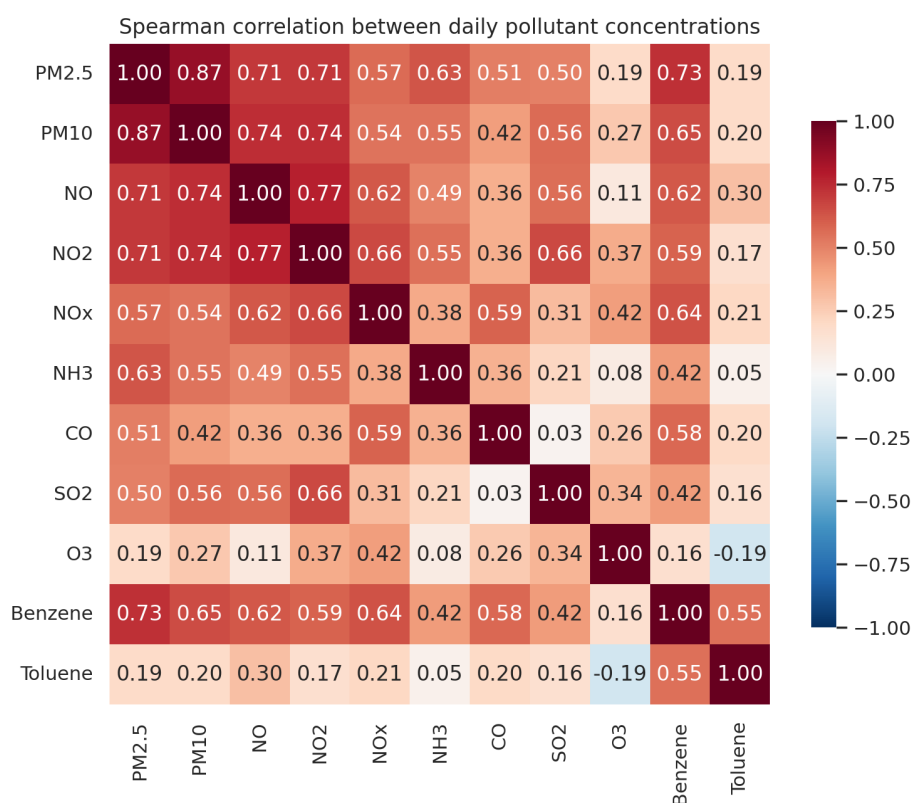


Figure 4. Spearman correlation between daily pollutant concentrations.

PM_{2.5} correlated most strongly with PM₁₀ (Spearman $r = 0.87$), benzene (0.73), NO and NO₂ (0.71 each), and NH₃ (0.63). O₃ and toluene were only weakly related to PM_{2.5} ($r = 0.19$ each). Three principal components had eigenvalues > 1 and together explained 75.5% of the variance (Table 2, Figures 5–6).

Table 2. PCA loadings (correlation between each pollutant and each component). |loading| > 0.6 considered strong.

Pollutant	PC1 (49.0%)	PC2 (14.1%)	PC3 (12.3%)
PM2.5	+0.90	+0.09	-0.16
PM10	+0.88	-0.05	-0.12
NO	+0.87	-0.01	+0.14
NO2	+0.88	-0.23	-0.05
NH3	+0.62	+0.28	-0.37
CO	+0.27	+0.65	-0.45
SO2	+0.62	-0.56	+0.27
O3	+0.35	-0.56	-0.36
Benzene	+0.82	+0.29	+0.25
Toluene	+0.38	+0.36	+0.75

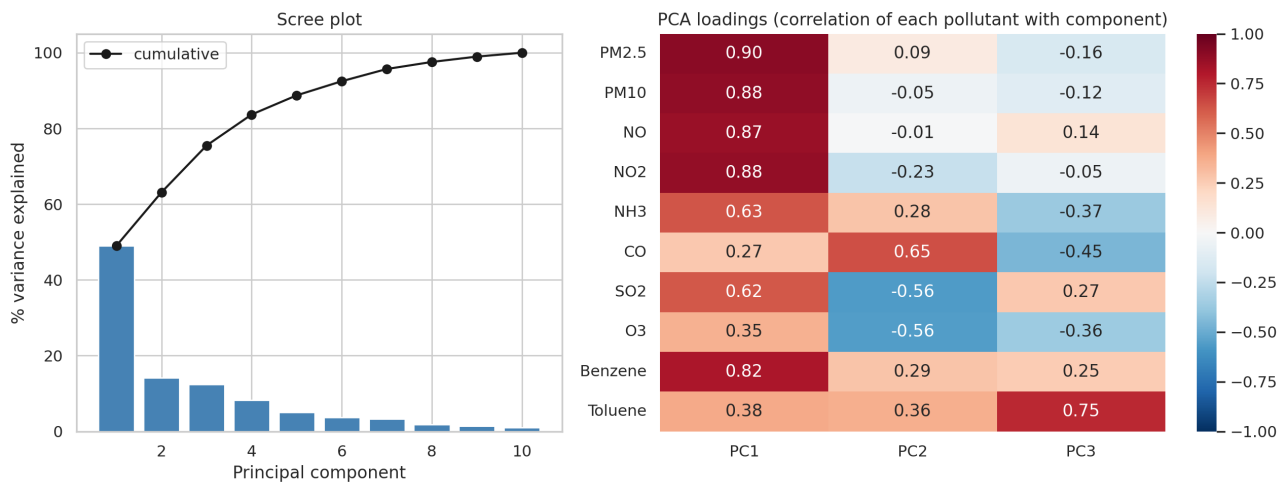


Figure 5. Scree plot (left) and component loadings (right).

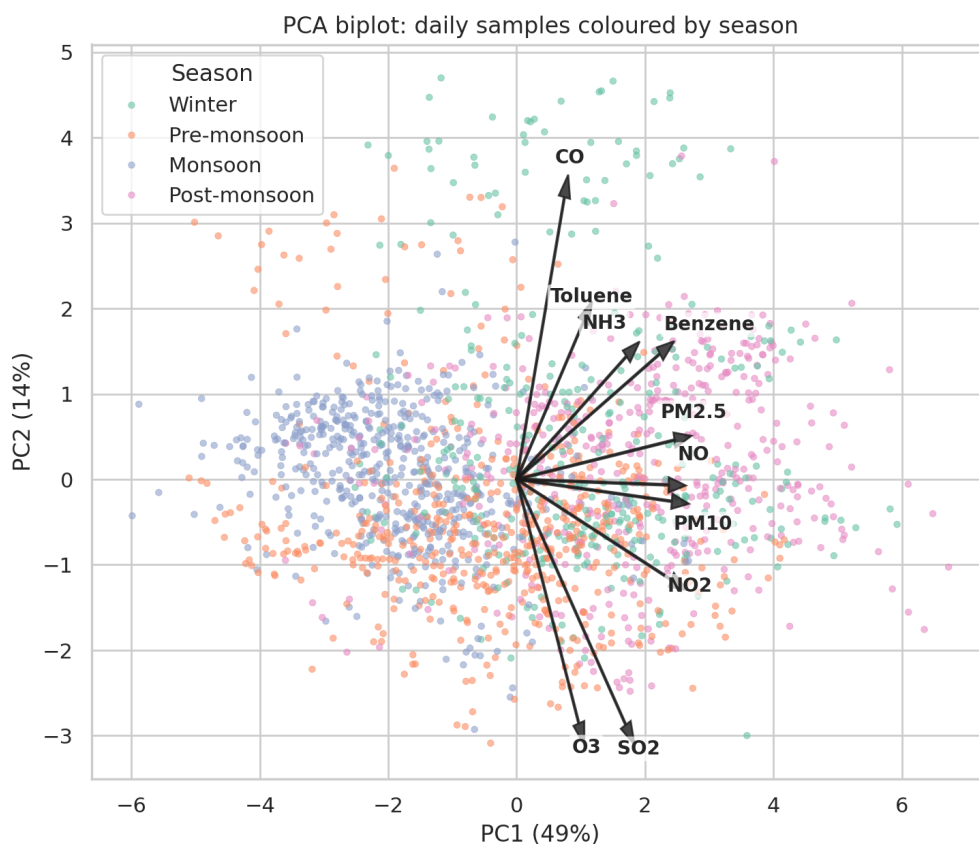


Figure 6. PCA biplot: daily samples in PC1–PC2 space coloured by season; arrows show pollutant loadings.

PC1 (49%) – common accumulation/combustion factor. PM_{2.5}, PM₁₀, NO, NO₂ and benzene all load strongly (0.82–0.90). Samples with high PC1 scores are almost entirely post-monsoon and winter days (Figure 6). This component combines two effects that this dataset cannot separate: shared combustion sources (traffic and other burning emit particles, NO_x and benzene together) and shared meteorology (shallow winter mixing layers raise all primary pollutants simultaneously).

PC2 (14%) – cold-season incomplete-combustion vs. photochemical contrast. CO loads positively (+0.65) together with weaker contributions from toluene, benzene and NH₃, while O₃ (–0.56) and SO₂ (–0.56) load negatively. Winter days sit highest on this axis, consistent with low-temperature and incomplete combustion (e.g. biomass and residential burning) in the coldest months, whereas ozone, being produced photochemically, is higher in the sunnier pre-monsoon period. The grouping of SO₂ with O₃ is not straightforward to interpret and would require station-level data to resolve.

PC3 (12%) – toluene/VOC factor. Toluene loads strongly (+0.75) and nearly independently of the other pollutants, suggesting a distinct volatile-organic source such as fuel evaporation or solvent use rather than the main combustion signal.

These components describe co-variation, not source contributions; they are hypotheses to be tested with speciated data and receptor modelling, not a source apportionment.

3.4 COVID-19 lockdown as a natural experiment

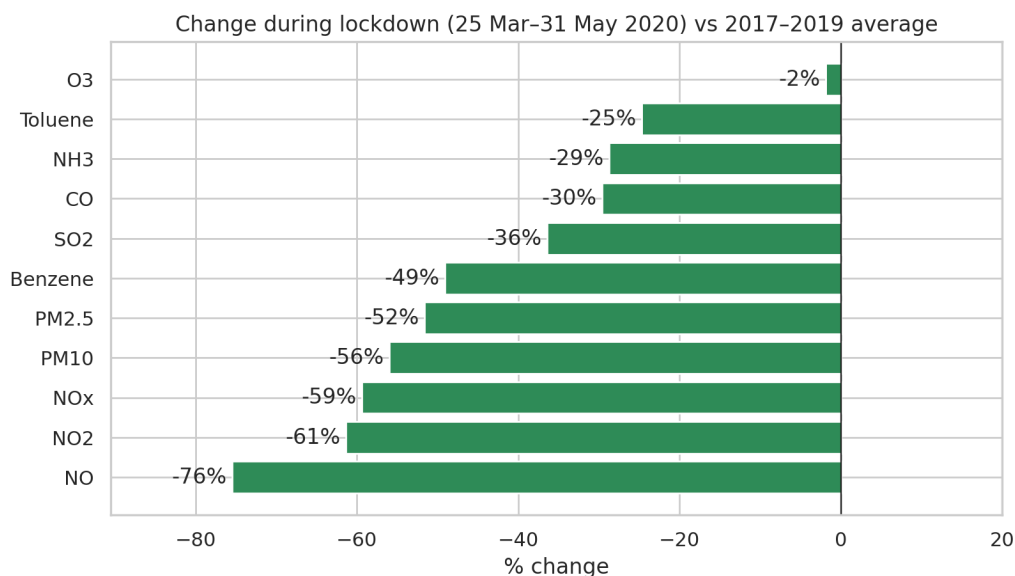


Figure 7. Change in mean concentration, 25 March–31 May 2020 vs. the 2017–2019 average for the same window.

Table 3. Mean concentrations during lockdown vs. baseline ($\mu\text{g}/\text{m}^3$; CO in mg/m^3).

Pollutant	2017–2019 mean	Lockdown 2020	Change
NO	39.7	9.7	-76%
NO2	58.1	22.4	-61%
PM10	261.7	115.1	-56%
PM2.5	100.1	48.5	-52%
Benzene	3.2	1.6	-49%
SO2	23.7	15.1	-36%
CO	1.2	0.8	-30%
NH3	40.2	28.7	-29%
Toluene	15.8	11.9	-25%
O3	50.6	49.7	-2%

The largest reductions were in NO (-76%) and NO₂ (-61%), which are dominated by vehicular exhaust, confirming that traffic is the main NO_x source in Delhi. PM_{2.5} (-52%), PM₁₀ (-56%) and benzene (-49%) also fell by about half. In contrast, O₃ was essentially unchanged (-2%) despite the large NO_x reduction. This is consistent with reduced titration of ozone by NO (less NO available to destroy O₃) and with non-linear ozone chemistry, and illustrates why cutting NO_x alone does not guarantee lower ozone. Part of the decrease may reflect meteorology in spring 2020, which this analysis does not control for.

3.5 Predicting daily PM_{2.5}

Table 4. Model performance on the 2019–2020 test period (n = 548 days). A = all co-pollutants incl. PM₁₀; B = gaseous pollutants only.

Feature set	Model	R ²	RMSE (µg/m ³)	MAE (µg/m ³)
Baseline	Monthly climatology	0.41	59.9	44.9
A	Linear Regression	0.89	26.3	17.5
A	Random Forest	0.94	19.3	14.3
B	Linear Regression	0.70	42.8	27.2
B	Random Forest	0.79	35.9	25.0

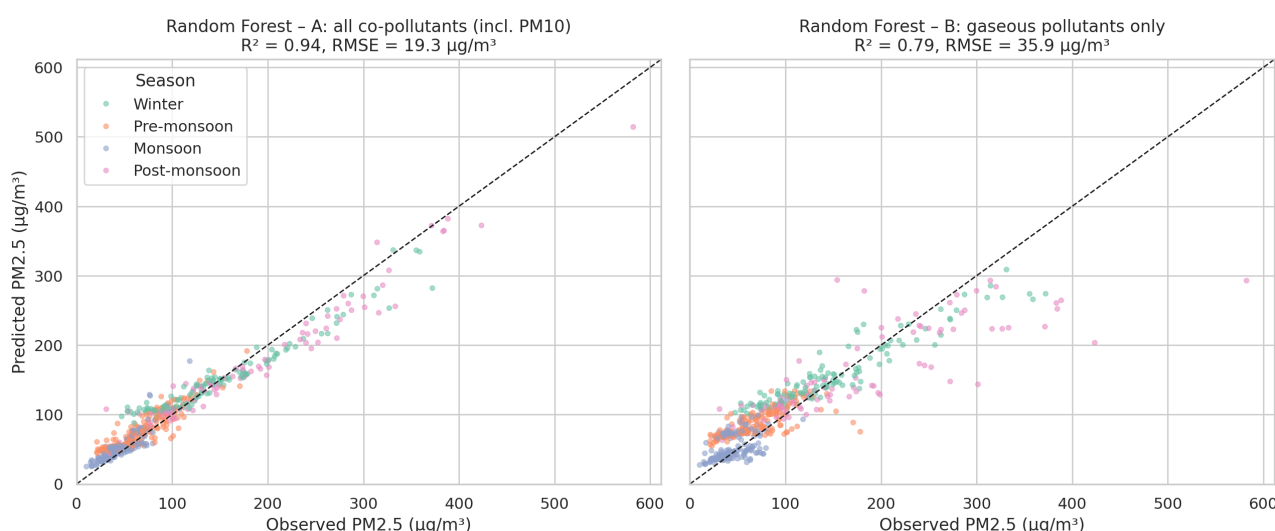


Figure 8. Predicted vs. observed PM_{2.5} on the test period for the random forest with feature sets A and B.

Seasonality alone (monthly climatology) explained 41% of test-period variance. Gaseous pollutants raised this to $R^2 = 0.79$ (random forest, RMSE 35.9 µg/m³), and including PM₁₀ raised it to $R^2 = 0.94$ (RMSE 19.3 µg/m³). The random forest outperformed linear regression in both cases, indicating non-linear relationships. Both models under-predicted the most extreme episodes (> 300 µg/m³, mostly November): tree-based models cannot extrapolate beyond the range seen in training, and such extreme days are rare.

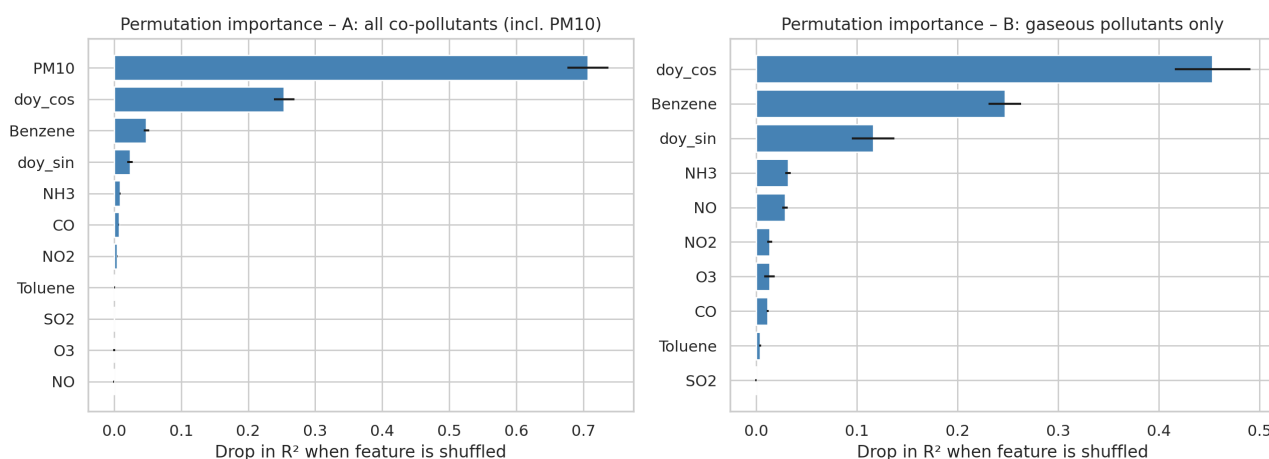


Figure 9. Permutation importance (drop in test R² when a feature is randomly shuffled).

Among gaseous pollutants, **benzene** was by far the most informative predictor of PM_{2.5}, followed by NH₃ and NO. Benzene is a widely used tracer of vehicular and combustion emissions, so this supports the PC1 interpretation. NH₃ is a precursor of secondary ammonium nitrate and sulphate aerosol, which forms a substantial part of winter PM_{2.5} in the Indo-Gangetic Plain. Seasonality terms remained the single largest contributor, reflecting the meteorological control that the model otherwise lacks.

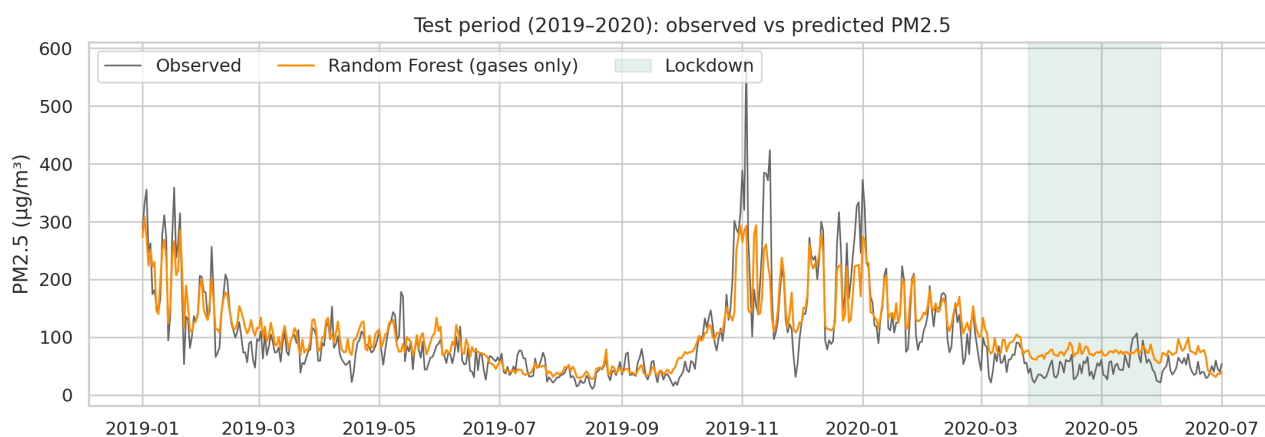


Figure 10. Observed and predicted (random forest, gaseous pollutants only) $PM_{2.5}$ over the test period; the lockdown window is shaded.

Outside the lockdown, the gas-only model's mean bias was $+4.6 \mu\text{g}/\text{m}^3$ (+4%). During the lockdown it rose to $+24.1 \mu\text{g}/\text{m}^3$ (+50% of the observed mean of $48.5 \mu\text{g}/\text{m}^3$). The model, having learned typical spring relationships between gases, season and particles, expected more $PM_{2.5}$ than was observed. This is a practical warning for data-driven air-quality models: they encode the emission regime they were trained on and can fail when that regime changes – precisely the situation in which policy-makers most want to evaluate interventions.

4. Limitations

- **City-averaged data** hide spatial differences between traffic, industrial and residential stations, and can produce artefacts (e.g. the $PM_{2.5} > PM_{10}$ days).
- **No meteorological variables** (wind speed and direction, boundary-layer height, temperature, humidity) were available, so emission and meteorological effects cannot be separated – in PCA, in the lockdown comparison, or in the prediction models.
- **PCA is not source apportionment.** Components reflect co-variation from any cause. Quantitative source contributions require speciated PM (ions, OC/EC, elements) and receptor models such as PMF.
- **Short record.** Five years is insufficient for robust trend detection without formal, deseasonalised trend tests.

5. Conclusions and future work

Routinely available CPCB data are sufficient to characterise the timing and severity of Delhi's $PM_{2.5}$ problem, to show that cold-season episodes are combustion-dominated rather than dust-dominated, and to identify benzene, NO_x and NH_3 as the gaseous pollutants most closely tied to $PM_{2.5}$. A random forest using only gaseous pollutants and seasonality explained 79% of daily $PM_{2.5}$ variance on unseen years, but degraded markedly during the lockdown, highlighting the dependence of data-driven models on the emission regime they are trained on. Natural next steps are:

- adding ERA5 or IMD meteorology to separate emission and weather effects;
- repeating the analysis at station level (CPCB CAAQMS) to contrast traffic-dominated and residential sites;
- linking post-monsoon peaks to satellite fire counts (NASA FIRMS) over Punjab and Haryana; and
- applying PMF to speciated $PM_{2.5}$ data for a true, quantitative source apportionment.

References

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